IGCSE 0607 Practice Questions — E1.9

Estimation

Total Questions: 12 — **Total Marks:** 60 Marks

Table of Contents & Quick Navigation

- **E1.9** Estimation
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

---

1 Consider the number 47.5381.

(a) Write 47.5381 correct to:

(i) 2 decimal places.

..... **[1]**

(ii) 2 significant figures.

..... **[1]**

(iii) the nearest integer.

..... **[1]**

2 **(a)** Write 0.04758 correct to 2 significant figures.

..... **[1]**

(b) Write 384.6 correct to the nearest 10.

..... **[1]**

(c) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.7 \times 4.12}{19.6 - 2.87}(0)$$

..... **[2]**

3 **(a)** Write 203.498 correct to

(i) 2 decimal places.

..... **[1]**

(ii) 3 significant figures.

..... **[1]**

(iii) the nearest ten.

..... **[1]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{6.89 \times 41.2}{0.513}$$

..... **[2]**

4 By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.4 \times 3.09}{0.187}$$

..... **[3]**

5 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{3.87 \times 10^6}}{0.0524 \times 21.3}$$

..... **[3]**

(b) Is your estimate in part (a) greater than or less than the exact value of the expression? Give a reason for your answer.

.....

..... **[2]**

6 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.1^3}{0.192 \times 48.7}$$

You must show all your working.

..... **[3]**

(b) Given that $y = \dfrac{ax}{b-c}$, find an estimate for the value of y when $a = 3.92$, $x = 0.0514$, $b = 1.08$ and $c = 0.812$.
Write all numbers correct to 1 significant figure in your calculation.

$y =$ ..... **[3]**

7 A rectangular field has length 125 m and width 74 m.

(a) Calculate the area of the field.
Give your answer correct to 2 significant figures.

..... m^2 **[2]**

(b) A different field is a square of side x metres.
The area of this square field is equal to the area of the rectangular field.
Calculate the value of x.
Give your answer correct to 1 decimal place.

$x =$ ..... **[3]**

8 A solid metal cone has radius $r = 5.82$ cm and height $h = 11.4$ cm.

(a) Calculate the volume of the cone.
Give your answer correct to 3 significant figures.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

..... cm^3 **[2]**

(b) By writing each of the values 5.82, 11.4 and π correct to 1 significant figure, estimate the volume of the cone.
Show your working.

..... cm^3 **[2]**

(c) A worker packs these cones into boxes.
Each box holds exactly 12 cones.
The factory produces 850 cones a day.
Calculate the maximum number of boxes that can be completely filled each day.

..... **[2]**

9 A solid metal cylinder has a base radius of 4.15 cm and a height of 19.8 cm. The metal has a density of 8.13 grams per cubic centimetre.

(a) By writing each number correct to 1 significant figure, estimate the mass of the cylinder in grams.
Use $\pi \approx 3$ for your estimation.
Show your working clearly.

..... g **[3]**

(b) Calculate the actual mass of the cylinder.
Give your answer correct to 3 significant figures.

..... g **[3]**

10 Write the following numbers correct to the specified degree of accuracy.

(a) 782.51 correct to the nearest ten.

..... **[1]**

(b) 0.03561 correct to 2 significant figures.

..... **[1]**

(c) 4.098 correct to 2 decimal places.

..... **[1]**

11 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{7.84 \times 31.6}{0.381}$$

..... **[2]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.6} + 19.3}{0.513^2}$$

..... **[3]**

12 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{38.7 \times 5.12^2}{\sqrt[3]{996}}$$

..... **[3]**

(b) A solid metal cylinder has radius 3.89 cm and height 11.2 cm.

By writing each dimension correct to 1 significant figure, estimate the volume of the cylinder.
Take the value of π to be 3.

..... cm^3 **[3]**

(c) A train travels a distance of 4210 km at an average speed of 585 km/h.

By writing both the distance and the speed correct to 1 significant figure, estimate the time taken for the journey.
Give your answer in hours and minutes.

hours, minutes **[3]**